A Universal Test-Scenario Roadmap for Open Research

Version B

Abstract

Open research often asks whether a proposed idea, method, model, or system “works.” But it does not always ask whether the testing plan itself was good enough. A project may pass some tests and fail others. A contributor may then conclude that the whole idea failed, when the real problem was that the test measured the wrong thing, used the wrong range, lacked a valid baseline, or skipped an important safety condition.

This paper proposes a **Universal Test-Scenario Roadmap**. It is a shared way to plan, document, and interpret tests across different fields. It helps researchers separate four very different outcomes:

- The idea itself failed.
- The test was poorly designed.
- The measurement was wrong or incomplete.
- The idea works only under certain conditions.

The roadmap is not meant to replace experts. It is meant to give open-research contributors a common set of rules so that results from different people, labs, sites, and fields can be compared fairly.

Core Idea

Every useful test should answer one clear question:

> “What exactly are we trying to learn, and what result would change our mind?”

A test should not simply ask whether something “works.” It should state:

- What condition is being tested.
- What will be changed.
- What will be measured.
- What result would count as success.
- What result would count as failure.

- What conditions would make the result invalid.
- What the result does **not** prove.

This prevents a common mistake: treating one failed test as proof that an entire idea is impossible.

1. Separate the Questions

A roadmap should separate different kinds of questions. A system may be possible but impractical. It may work at small scale but fail at large scale. It may work under one condition but fail under another.

Question	Plain-language test	Valid conclusion
Can this happen at all?	Feasibility test	“Possible,” “not observed,” or “not demonstrated”
Does it work better than the current method?	Comparison test	“Better,” “not better,” or “unclear”
Which conditions matter most?	Screening test	“This factor matters,” “this factor does not matter here,” or “it depends on another factor”
Can it handle stress or change?	Recovery or robustness test	“Stable,” “limited,” or “unsafe outside this range”
Can someone else repeat it?	Replication test	“Reproducible,” “partly reproducible,” or “not reproducible”
Can it handle more volume, load, or demand?	Scale test	“Scales,” “degrades,” or “needs redesign”

For example, a plant may fail to grow under high nutrient concentration but grow well at a lower concentration. A website may fail under heavy traffic but work well under normal traffic. A material may crack under sudden heating but remain stable when heated slowly. In each case, the result is useful only if the test conditions are recorded.

2. The Dry Human-in-the-Loop Flywheel

The roadmap uses a simple industrial idea: before testing something in the real world, test it in a safe, reversible, or simulated setting. This is called a **dry human-in-the-loop flywheel**.

“Dry” means the test does not immediately affect a live system, patient, crop, machine, community, or environment. “Human-in-the-loop” means people review the plan, inspect the evidence, approve changes, and retain final authority.

The flywheel has five repeating steps:

text

```
Plan
   ↓
Run a small, safe test
   ↓
Observe and check the result
   ↓
Human review
   ↓
Update the plan, model, or roadmap
   ↓
Repeat
```

In plain language:

1. **Plan:** State what will be tested, what will be measured, and what result would count as success.
2. **Run a small test:** Change only what is necessary, and keep the change small and reversible.
3. **Observe:** Record what happened, including anything unexpected.
4. **Check:** Compare the result with the predefined success and failure criteria.
5. **Review and update:** Decide whether to continue, repeat, revise, pause, or stop.

This mirrors established industrial practice. In construction and infrastructure, constructability reviews bring together designers, builders, maintenance staff, and field representatives so that practical site knowledge can improve a design before and during construction. In manufacturing and operations, the Plan–Do–Check–Act cycle uses a similar loop: plan a change, test it, check the result, and either standardize or revise it.

3. Four Universal Test Scenarios

These four scenarios can be adapted to almost any field.

Scenario 1: Can It Happen at All?

Purpose: Find out whether the basic idea is possible under carefully controlled conditions.

Plain-language procedure:

1. State the minimum condition needed for the idea to make sense.
2. Set up a baseline or comparison condition.
3. Apply the smallest change needed to test the idea.
4. Measure whether the expected outcome appears.
5. Record the result without overstating it.

Examples:

- **Biology:** Does a plant tissue sample form new growth under a defined nutrient and light condition?
- **Chemistry:** Does a catalyst produce the intended product under a defined temperature and mixture?
- **Software:** Does a new algorithm produce a correct result on a simple test case?
- **Materials:** Does a new material form the intended structure under a defined heating process?
- **Civil engineering:** Can a proposed foundation method be built under known soil and site conditions?

Important rule:
If the expected outcome does not appear, that does **not** prove the whole idea is impossible. It only means it was not demonstrated under the conditions tested.

Scenario 2: Can It Recover from a Small Shock?

Purpose: Find out whether a system can handle a small, controlled disturbance and return to a safe or normal state.

Plain-language procedure:

1. Establish a stable baseline.
2. Make one small, reversible change.
3. Watch how far the system moves from normal.
4. Watch how long it takes to return.
5. Only increase the change if the system recovered safely.

Examples:

- **Bioprocessing:** Add a small amount of nutrient and watch whether oxygen levels recover.
- **Thermal systems:** Raise temperature slightly and see how long the system takes to stabilize.
- **Software:** Send a short burst of extra requests and see whether response time returns to normal.

- **Power systems:** Add a small load and see whether voltage or frequency remains stable.
- **Field engineering:** Apply a small load or flow change and watch whether the structure or equipment responds safely.

Important rule:
A failed recovery does not mean the system is useless. It means the tested level is too high, too fast, or outside the safe range. The correct response is to return to the last stable level and revise the test.

Scenario 3: Which Factors Actually Matter?

Purpose: Find out which inputs matter most and whether some inputs only matter when combined with others.

Plain-language procedure:

1. List the possible factors that could affect the result.
2. Choose a low and high level for each factor.
3. Test combinations systematically rather than changing everything at once.
4. Record which factors have the largest effect.
5. Identify whether two factors interact.

This is called **screening**. Screening designs are used when many possible factors could matter, and the goal is to identify the few that most influence the outcome.itl.nist+2

Examples:

- **Biology:** Test nutrient mix, light color, temperature, and cell density.
- **Chemistry:** Test catalyst amount, temperature, mixing time, and ingredient ratio.
- **Manufacturing:** Test pressure, temperature, material type, and processing time.
- **Software:** Test data size, model settings, hardware type, and preprocessing method.
- **Agriculture:** Test water, fertilizer, planting density, and light exposure.

Important rule:
A factor that does not matter in one range may matter in another. The result should say, “This factor had little effect within the tested range,” not “This factor never matters.”

Scenario 4: Can We Predict the Best Safe Range?

Purpose: Once the important factors are known, map how they affect the outcome and find a safe operating range.

Plain-language procedure:

1. Select the few factors that mattered most in screening.
2. Test several levels of those factors.
3. Record the outcome for each combination.
4. Use the results to build a simple prediction model.
5. Predict a promising operating condition.
6. Run a new confirmation test to see whether the prediction holds.

This is called **response-surface testing**. It is used to understand how several factors affect an outcome, including cases where the relationship is curved or where two factors interact. After a model predicts a good setting, confirmation runs are needed to verify that the prediction actually works.itl.nist+2

Examples:

- **Bioprocessing:** Map nutrient feed, temperature, and pH against growth or product yield.
- **Chemistry:** Map catalyst amount, temperature, and ingredient ratio against product yield.
- **Materials:** Map curing temperature, pressure, and composition against strength.
- **Software:** Map user load, batch size, and cache size against speed and errors.
- **Agriculture:** Map water, fertilizer, and planting density against crop yield.

Important rule:
A prediction is not proof. The predicted condition must be tested again before it can be called validated.osti+2

4. Volume Dial-In: Growing Capacity Safely

Many projects fail during scale-up because they jump from a small successful test to a much larger one. A better approach is to increase volume, load, or demand gradually while watching how the system responds.

Phase 1: Establish the Baseline

Before increasing anything, prove that the system is stable at its current level.

- Record normal output, error rate, resource use, and environmental conditions.
- Repeat the baseline test enough times to understand normal variation.
- Define what “recovery” means.
- Define the maximum allowed deviation from normal.

Phase 2: Apply a Small Pulse

Increase volume or demand by a small, reversible amount.

- Do not make a permanent jump.
- Watch the immediate response.
- Record the highest deviation, how long it lasted, and how long recovery took.
- Return to baseline before trying again.

This follows the logic of pulse-feeding studies, where a small controlled addition is used to see whether the system can absorb it safely.

Phase 3: Increase in Small Steps

If the pulse was handled safely, increase volume in small steps.

- Use a fixed increase, such as 5% or 10%.
- Stay at each level long enough to see both the immediate response and the longer-term behavior.
- Require repeated success before moving higher.
- If a level fails, return to the last stable level.

Stepwise feeding has been used in bioprocessing to give a system time to adapt between increases, rather than forcing an abrupt transition. Pilot-scale work has also used controlled feed-rate progression while monitoring oxygen to avoid overload.

Phase 4: Build a Simple Prediction Model

Use the results from each successful level to predict what will happen at the next level.

In plain language:

> "Based on what happened at the previous levels, what do we expect to happen if we increase volume slightly?"

The model should predict:

- How far the system will move from normal.
- How long recovery will take.
- Whether any safety limit may be approached.
- How much uncertainty exists in the prediction.

The model should only be used within the range that was actually tested. It should not be used to claim that the system will work far beyond that range.

Phase 5: Define Safe Zones

Translate the model into clear operating zones.

Zone	Meaning	Action
Approved	The system stays within tested safe limits	Continue normal operation
Caution	The system works but is approaching a limit	Slow down, monitor more closely, or require approval
Restricted	The system may exceed a safety limit	Stop increasing or return to the last stable level
Unvalidated	The level has not been tested	Do not claim it works

Phase 6: Confirm Before Locking In

Before declaring a higher-volume level successful, repeat it.

- Run the same condition more than once.
- Check that the result matches the prediction.
- Confirm that no safety or recovery limit was violated.
- Record the validated level and its limits.
- Only then treat it as the new baseline.

Response-surface methods commonly require confirmation experiments after a model predicts a promising setting. These confirmation runs test whether the prediction can be reproduced under real conditions.itl.nist+2

5. Feasibility Gates: What Must Be True First

The first phase should contain only **minimum requirements**. These are not performance goals. They are conditions that must be true before later tests can mean anything.

Examples:

- The measuring tool works and is calibrated.
- The baseline is stable.
- The change can be made safely and reversibly.

- The outcome can be measured clearly.
- Safety limits are defined before testing.
- There is a control, reference, or comparison condition.
- The data includes time, conditions, and who or what performed the test.
- The test can be repeated by someone else.

If a feasibility gate fails, the correct response is not “the idea failed.” The correct response is:

- Fix the instrument.
- Stabilize the baseline.
- Redefine the measurement.
- Reduce the change.
- Change the test range.
- Redesign the test.
- Mark the result as inconclusive.

6. Rules of Engagement for Open Research

To make contributions comparable, every test should be recorded in the same basic format.

Field	What to record
Claim being tested	The exact idea or capability being evaluated
Test type	Feasibility, comparison, recovery, screening, prediction, scale, or replication
What was changed	The input variable and how much it changed
What was kept constant	The conditions held steady during the test
Baseline	The reference condition used for comparison
What was measured	The outcome and how it was measured
Success rule	What result counts as a pass
Failure rule	What result counts as a stop or failure
Uncertainty	Measurement error, variation, or confidence range
Result	Pass, fail, unclear, conditional, or not demonstrated
What it proves	The limited conclusion supported by the result
Next step	Continue, repeat, revise, pause, or stop

This prevents vague reports such as “it did not work.” A useful report explains what was tested, under what conditions, how it was measured, and what the result does not prove.

7. Result Categories

Every result should be placed into one of six categories.

Category	Meaning	Next action
Validated	The predefined success rule was met	Continue, replicate, or expand carefully
Conditional	It works only under stated conditions	Define the boundaries
Unclear	The test could not tell success from failure	Improve the test or measurement
Not demonstrated	The expected result did not occur under those conditions	Do not generalize beyond the test
Restricted	It works only with lower load or added safeguards	Define limits and safeguards
Rejected for this pathway	Evidence supports stopping this specific approach	Document why and preserve the evidence

This prevents binary thinking. A result is not simply “working” or “not working.” It is a bounded statement about what was tested, under what conditions, and with what uncertainty.

8. Common Roadmap Failures

Problem	What goes wrong	Roadmap correction
Stage confusion	A scale failure is treated as proof the idea is impossible	Separate feasibility, performance, and scale tests
Wrong measurement	The test does not measure the claimed mechanism	Define the outcome before testing
No baseline	There is nothing valid to compare against	Require baseline validation
Too many changes	Several variables change at once	Use screening or controlled test designs

Overconfident prediction	A model is treated as proof	Require confirmation runs
Extrapolation	A result is applied outside the tested range	Restrict conclusions to tested conditions
Hidden failure	The system seems to work, but the measurement was invalid	Require sensor checks and controls
Only successes reported	Failed tests disappear	Document negative and unclear results
No stopping rule	Contributors keep testing without a decision point	Define pass, fail, hold, and escalation rules
Design-field separation	Field observations never reach the people revising the design	Use the human-in-the-loop flywheel

9. Open-Research Contribution Model

The roadmap should be a living document. Contributors should be able to add:

- New test scenarios.
- New measurements.
- New failure modes.
- Replication results.
- Negative results.
- Unclear results.
- Better safety limits.
- Improved prediction models.
- Evidence that a proposed limit is too strict or too loose.

Each contribution should include the minimum test record, the result category, and a clear statement of what the result does and does not prove.

A useful open-research rule is:

> **No result should be interpreted outside the stage, conditions, measurements, and uncertainty under which it was obtained.**

This allows contributors to disagree productively. One person may show that a method works under one condition; another may show that it fails under another. Both can be valid if the roadmap records the conditions and prevents unfair generalization.

10. Discussion

This framework does not claim that every scientific question can be tested the same way. A biology experiment, a materials test, a software benchmark, and a community field trial require different tools, controls, and safety rules.

What can be universal is the **structure of the roadmap**:

1. Define what must be true before testing.
2. Make one controlled change at a time.
3. Measure a predefined outcome.
4. Separate immediate response from long-term behavior.
5. Use predictions only within tested ranges.
6. Confirm predictions with new tests.
7. Record uncertainty.
8. Preserve negative and unclear results.
9. Update the roadmap when a test reveals a missing condition.
10. Prevent conclusions from crossing stages without evidence.

The industrial dry human-in-the-loop flywheel provides the bridge. In infrastructure practice, constructability reviews bring field knowledge into design before construction begins, and field representatives continue providing feedback during construction so that repeated problems can be reduced. Open research can use the same principle: theoretical contributors, experimental contributors, and domain practitioners should work in one shared learning loop.

11. Conclusion

A research roadmap should not merely list experiments. It should define the conditions under which each experiment can produce a meaningful conclusion.

The Universal Test-Scenario Roadmap separates feasibility, performance, robustness, prediction, scale, and replication into clear stages. It uses small controlled tests, recovery checks, screening methods, prediction models, and confirmation runs to prevent both premature rejection and premature acceptance.

Its central rules are simple:

- Prove the basics before testing performance.
- Make one controlled change at a time.
- Watch recovery before increasing load.
- Use predictions only within tested ranges.
- Confirm predictions with new tests.
- Record uncertainty and negative results.
- Keep humans responsible for safety, interpretation, and final decisions.

- Improve the roadmap whenever evidence shows that something was missed.

By making the testing structure explicit, open-research projects can distinguish between a failed idea, a failed test, a failed measurement, and an incomplete roadmap. That distinction is necessary for cumulative, reproducible, and trustworthy open science.

Optional Technical Appendix

The main paper should remain equation-free. A short appendix can provide the technical versions for specialists:

- **Baseline variation:** average, spread, and repeat-run variability.
- **Recovery test:** peak deviation, time to peak, recovery time, and residual error.
- **Screening design:** factorial, fractional factorial, or Plackett–Burman designs.
- **Prediction model:** linear, interaction, or curved response-surface models.
- **Confirmation rule:** repeated runs at a predicted setting to verify that the prediction holds.itl.nist+2

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